

Canonical Correlation Analysis

Health and Lifestyle Dataset

Examining the Relationship Between Lifestyle Behaviors and Health Outcomes

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1. Introduction and Study Context

This analysis investigates the relationship between lifestyle behavioral patterns and physiological health outcomes using Canonical Correlation Analysis (CCA). The dataset contains **100 subjects** measured on six variables, partitioned into two sets: a **lifestyle set (X)** with three behavioral indicators, and a **health outcomes set (Y)** with three physiological measures.

Variable Sets

Set	Variables	Description	Count
X — Lifestyle	Exercise, DietScore, SleepHours	Behavioral/lifestyle indicators	p = 3
Y — Health	BloodPressure, Cholesterol, BMI	Physiological health outcomes	q = 3

With $\min(p, q) = \min(3, 3) = 3$ **canonical variate pairs**, CCA will identify up to three independent dimensions of association between lifestyle behaviors and health outcomes.

2. Descriptive Statistics

Variable	Set	Mean	Std Dev	Role
Exercise	Lifestyle	4.540	1.755	Exercise frequency
DietScore	Lifestyle	6.660	1.719	Diet quality score
SleepHours	Lifestyle	6.560	1.157	Sleep hours/night
BloodPressure	Health	126.320	5.747	Systolic BP (mmHg)
Cholesterol	Health	197.160	9.768	Total cholesterol (mg/dL)
BMI	Health	22.258	2.636	Body mass index

Subjects averaged about 4.5 exercise sessions, a diet score of 6.7, and 6.6 hours of sleep per night. Health-wise, the average systolic blood pressure was 126.3 mmHg, cholesterol was 197.2 mg/dL, and BMI was 22.3 — all near healthy reference ranges, with meaningful individual variation captured by the standard deviations.

3. Canonical Correlations

The three canonical correlations and their squared values are shown below. r^2 represents the proportion of shared variance between each canonical variate pair.

Pair	Canonical Corr (r^*)	r^{*2}	Eigenvalue	Interpretation
r^*1	0.984667	0.969568	0.969568	Very Strong
r^*2	0.871236	0.759053	0.759053	Strong
r^*3	0.036385	0.001324	0.001324	Negligible

$r^2 = 0.9847$ ($r^2 = 0.9696$): This is an exceptionally strong canonical correlation. The first pair of canonical variates shares 97.0% of variance, indicating a very powerful relationship between the dominant lifestyle factor and the dominant health outcome factor.

$r^2 = 0.8712$ ($r^2 = 0.7591$): The second canonical correlation is also strong, capturing an additional 75.9% of shared variance along a second, independent dimension of the lifestyle–health relationship.

$r^2 = 0.0364$ ($r^2 = 0.0013$): The third correlation is negligible (0.13% shared variance) and will be confirmed as non-significant by the Wilks' Lambda tests.

4. Significance Tests — Wilks' Lambda

Wilks' Lambda tests each successive set of roots jointly. The Bartlett chi-square approximation uses multiplier $t = n - 1 - (p+q+1)/2 = 95.5$.

Test (H_0)	Wilks' Λ	Chi-Square	df	p-value	Decision
Roots 1–3 = 0	0.007323	469.552	9	0.000000	Reject H_0 (***)
Roots 2–3 = 0	0.240628	136.040	4	0.000000	Reject H_0 (***)
Roots 3–3 = 0	0.998676	0.127	1	0.722079	Fail to Reject

Roots 1–3 (all roots): $\Lambda = 0.0073$, $\chi^2(9) = 469.552$, $p < 0.001$

The overall multivariate relationship between lifestyle behaviors and health outcomes is highly significant. There is overwhelming evidence that the two variable sets are related.

Roots 2–3: $\Lambda = 0.2406$, $\chi^2(4) = 136.040$, $p < 0.001$

Even after the first canonical dimension is removed, the remaining relationship is still highly significant. A second independent lifestyle–health dimension exists.

Root 3 only: $\Lambda = 0.9987$, $\chi^2(1) = 0.127$, $p = 0.7221$

The third root is not significant ($p > 0.05$). Only two canonical dimensions carry meaningful information; the third should be discarded from interpretation.

5. Canonical Coefficients

5.1 Raw Canonical Coefficients — X Set (a vectors)

Raw coefficients form the canonical variates from original (unstandardized) variables. They are scale-dependent and should not be compared across variables directly.

Variable	U1	U2	U3
Exercise	0.4265	0.3764	0.0946
DietScore	0.3897	-0.4345	-0.0464
SleepHours	0.0260	-0.0143	-0.8792

5.2 Raw Canonical Coefficients — Y Set (b vectors)

Variable	V1	V2	V3
BloodPressure	-0.6329	0.2461	0.7341
Cholesterol	-0.2291	-0.9652	0.1260
BMI	-0.7395	0.0884	-0.6673

5.3 Standardized Canonical Coefficients — X Set

Standardized coefficients allow direct comparison of variable importance within each canonical variate. Larger absolute values indicate stronger contribution.

Variable	U1	U2	U3
Exercise	0.7485	0.6605	0.1659
DietScore	0.6698	-0.7467	-0.0797
SleepHours	0.0301	-0.0165	-1.0176

For U¹: Exercise (0.7485) and DietScore (0.6698) are the dominant contributors to the first lifestyle canonical variate, while SleepHours (0.0301) contributes minimally. For U²: DietScore reverses sign and dominates, with Exercise remaining positive — this second dimension contrasts diet versus exercise patterns. SleepHours dominates U³ exclusively.

5.4 Standardized Canonical Coefficients — Y Set

Variable	V1	V2	V3
BloodPressure	-3.6373	1.4143	4.2187
Cholesterol	-2.2381	-9.4278	1.2310
BMI	-1.9497	0.2332	-1.7592

All three health variables contribute negatively to V¹ with large coefficients, forming a composite 'poor health' dimension. For V², Cholesterol dominates with a large negative coefficient, contrasting with BloodPressure being positive — this dimension separates cholesterol from blood pressure patterns.

6. Canonical Structure Coefficients

Structure coefficients (canonical loadings) are correlations between original variables and canonical variates. They indicate how much each variable 'loads onto' each dimension. Loadings with $|r| \geq 0.30$ are considered substantively meaningful.

6.1 X Variables with Canonical Variates U₁–U₃

Variable	U ₁	U ₂	U ₃
Exercise	0.7449	0.6670	0.0112
DietScore	0.6573	-0.7529	0.0317
SleepHours	0.0700	0.1677	-0.9833

Exercise (U₁ loading = 0.7449) and DietScore (0.6573) load strongly and positively on U₁. SleepHours loads minimally on U₁ and U₂ but dominates U₃ (-0.9833), confirming that the three lifestyle variables contribute to three distinct canonical dimensions.

6.2 Y Variables with Canonical Variates V₁–V₃

Variable	V ₁	V ₂	V ₃
BloodPressure	-0.9527	-0.5382	0.9658
Cholesterol	-0.8379	-0.9897	0.7609
BMI	-0.9525	-0.6345	0.8259

All three health variables load very strongly (negatively) on V₁ (BloodPressure: -0.9527, Cholesterol: -0.8379, BMI: -0.9525). This confirms that V₁ captures an overall poor-health composite. On V₂, Cholesterol loads most strongly (-0.9897), distinguishing it from the others. V₃ is non-significant and should not be interpreted.

7. Redundancy Analysis

Redundancy indices measure how much variance in one variable set is explained by the canonical variates of the opposite set. This extends CCA by quantifying predictive usefulness.

Pair	Redundancy (X explained by V)	Redundancy (Y explained by U)
Pair 1	0.320581 (32.06%)	0.813407 (81.34%)
Pair 2	0.263131 (26.31%)	0.422982 (42.30%)
Pair 3	0.000427 (0.04%)	0.000968 (0.10%)

The health outcomes (Y) are explained remarkably well by the lifestyle canonical variates: Pair 1 alone accounts for 81.3% of Y-set variance, and Pair 2 adds another 42.3%. Together, Pairs 1 and 2 explain 123.6% of total health outcome variance through lifestyle canonical variates — a very high redundancy indicating strong predictive power. Conversely, the lifestyle variables are somewhat less explained by the health variates (58.4% combined), which is typical since lifestyle behaviors have more independent variation.

8. Overall Interpretation and Conclusions

1. Two significant canonical dimensions:

Canonical pairs 1 and 2 are both highly statistically significant ($p < 0.001$). The third pair is not significant ($p = 0.72$) and is discarded. This means lifestyle behaviors relate to health outcomes through two independent channels.

2. First canonical dimension ($r^2 = 0.9847$):

This dimension represents an overall healthy lifestyle → good health outcomes axis. Higher Exercise and DietScore jointly predict lower BloodPressure, Cholesterol, and BMI. This is the dominant pathway — it accounts for the vast majority of the shared variance between the two variable sets.

3. Second canonical dimension ($r^2 = 0.8712$):

This dimension is also significant and represents a contrast between DietScore and Exercise: higher diet quality (with lower exercise) is associated with lower Cholesterol specifically. This second dimension adds independent information about the diet–cholesterol pathway that is not captured by the first dimension.

4. SleepHours operates independently:

SleepHours loads primarily on U_3 (the non-significant third dimension), indicating that sleep has minimal association with blood pressure, cholesterol, and BMI in this dataset — at least independently from exercise and diet effects. Its contribution to the two significant dimensions is near zero.

5. Health outcomes are strongly predicted by lifestyle:

The combined redundancy of Y given U is 123.6%, meaning that the two significant lifestyle canonical variates collectively account for over 124% of the variance in the health outcome set. This is a remarkably strong predictive relationship.

6. Practical implication:

Interventions targeting both exercise frequency and diet quality simultaneously would have the broadest impact on all three health outcomes (blood pressure, cholesterol, and BMI). Diet quality specifically has an additional independent effect on cholesterol beyond its joint role with exercise.